



OPERATING MANUAL

Firmware Version 2.5.4 • Document v1.1

Voice Control Your Ambiance • App Control • Smart Scheduling • Over-the-Air Updates

Thank you for choosing Moonlight. This guide will help you set up, personalize, and enjoy your smart RGB lighting experience with seamless Alexa and Google integration.

Welcome to Moonlight

Moonlight transforms any space with beautiful, customizable RGB lighting that responds to your voice and adapts to your lifestyle. Whether you're creating the perfect movie night ambiance, waking up gently with sunrise simulation, or setting the mood for a dinner party, Moonlight puts elegant control at your fingertips — or simply by speaking.



Figure 0: Moonlight in a modern bedroom setting — voice-activated ambiance made simple and elegant.

Key Customer Benefits

- **Alexa & Google Compatible** — Control power, brightness, and full color with natural voice commands.
- **Beautiful Web App** — Modern mobile-friendly interface for precise color selection, 5 favorites, brightness, and multiple animation modes.
- **Smart Scheduling** — Set custom on/off times or let Moonlight automatically follow local sunrise and sunset.
- **Dynamic Animations** — Choose from smooth continuous scrolling or gentle timed color changes (1 min to 1 day intervals).
- **Future-Proof** — Automatic or one-click over-the-air firmware updates keep improving your lamp.
- **Instant Favorites** — Save and recall your perfect lighting scenes with a single tap.

This operating manual contains everything you need to get started quickly and make the most of your Moonlight experience. We recommend beginning with the Initial Setup section on the next page.

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1. Introduction and System Overview

The MoonLight RGB Lighting Controller is a sophisticated, network-enabled device designed to provide precise control over RGB LED lighting installations. Built upon the ESP32 microcontroller platform, it combines local web-based management, cloud voice assistant integration, intelligent scheduling, and seamless over-the-air firmware updates into a single, user-friendly package.

Key capabilities include real-time color and brightness adjustment via an intuitive web interface, multiple animation modes (static, smooth scrolling, and timed color transitions), five user-defined favorite color presets, time-based or solar-synchronized automation, and secure integration with Amazon Alexa and Google Home through the Sinric Pro platform.

All configuration data, including Wi-Fi credentials, color values, scheduling parameters, and cloud credentials, are persistently stored in the ESP32's non-volatile flash memory, ensuring settings survive power cycles and enabling fully autonomous operation once configured.

1.1 Intended Use

This device is intended for indoor residential, commercial, or light industrial RGB LED lighting control. It is suitable for accent lighting, mood lighting, architectural installations, and automated ambiance systems. The controller assumes the presence of appropriate LED drivers or constant-current power supplies matched to the connected LED load.

2. Hardware and Connectivity

2.1 Core Components

- **Microcontroller:** ESP32 (dual-core, Wi-Fi 802.11 b/g/n, Bluetooth LE)
- **LED Outputs:** Three independent PWM channels (GPIO 3 = Red, GPIO 4 = Green, GPIO 2 = Blue)
- **Storage:** 4 MB flash with NVS (Non-Volatile Storage) partition for Preferences
- **Networking:** 2.4 GHz Wi-Fi station and soft-AP modes; mDNS responder
- **Serial Console:** 115200 baud, 8N1 (optional verbose debug logging)

2.2 Network Architecture

The device operates in one of two mutually exclusive network modes:

- **Access Point (AP) Mode:** Activated automatically on first boot or when no home Wi-Fi credentials are stored. Broadcasts an open Wi-Fi network (SSID: MoonLight_Setup or MoonLight_MLxxxx). Web interface available at 192.168.4.1.
- **Station Mode:** Connects to an existing 2.4 GHz home or office Wi-Fi network. Web interface reachable via DHCP-assigned IP address or via mDNS (http://MoonLight_SN.local).

Note: The ESP32 Wi-Fi stack supports only the 2.4 GHz band. 5 GHz networks are not compatible.

3. Initial Setup Procedure

3.1 Power-On and AP Mode Entry

Apply power to the controller. If no valid home Wi-Fi credentials exist in flash memory, the device automatically enters Access Point Mode. The RGB LEDs will blink blue at approximately 1 Hz to visually confirm AP Mode operation.

3.2 Accessing the Web Interface (First-Time Setup)

1. Using a Wi-Fi-enabled device (phone, tablet, or laptop), scan for and connect to the MoonLight access point (SSID starts with “MoonLight_”).
2. Open any web browser and enter the URL **http://192.168.4.1** in the address bar.
3. The setup interface will load. Follow the on-screen prompts to connect the lamp to your home Wi-Fi network.

3.3 Configuring Home Wi-Fi Credentials

From the web interface (either in AP Mode or after initial connection):

1. Navigate to the **Settings** tab → **WiFi** sub-tab.
2. Enter the exact SSID (case-sensitive) of your 2.4 GHz home network.
3. Enter the Wi-Fi password (leave blank for open networks).
4. Click **Connect** (or **Switch Network** if already configured).
5. The device will reboot and attempt connection. Successful association is indicated by a green LED blink sequence followed by transition to NORMAL operating state.

If the connection attempt fails after approximately 30 seconds, the device automatically reverts to AP Mode to allow correction of credentials.

4. Web User Interface — Control Tab

The Control tab provides direct, real-time manipulation of the connected RGB LEDs. All changes are applied immediately and persisted to flash memory after a short delay (background save).

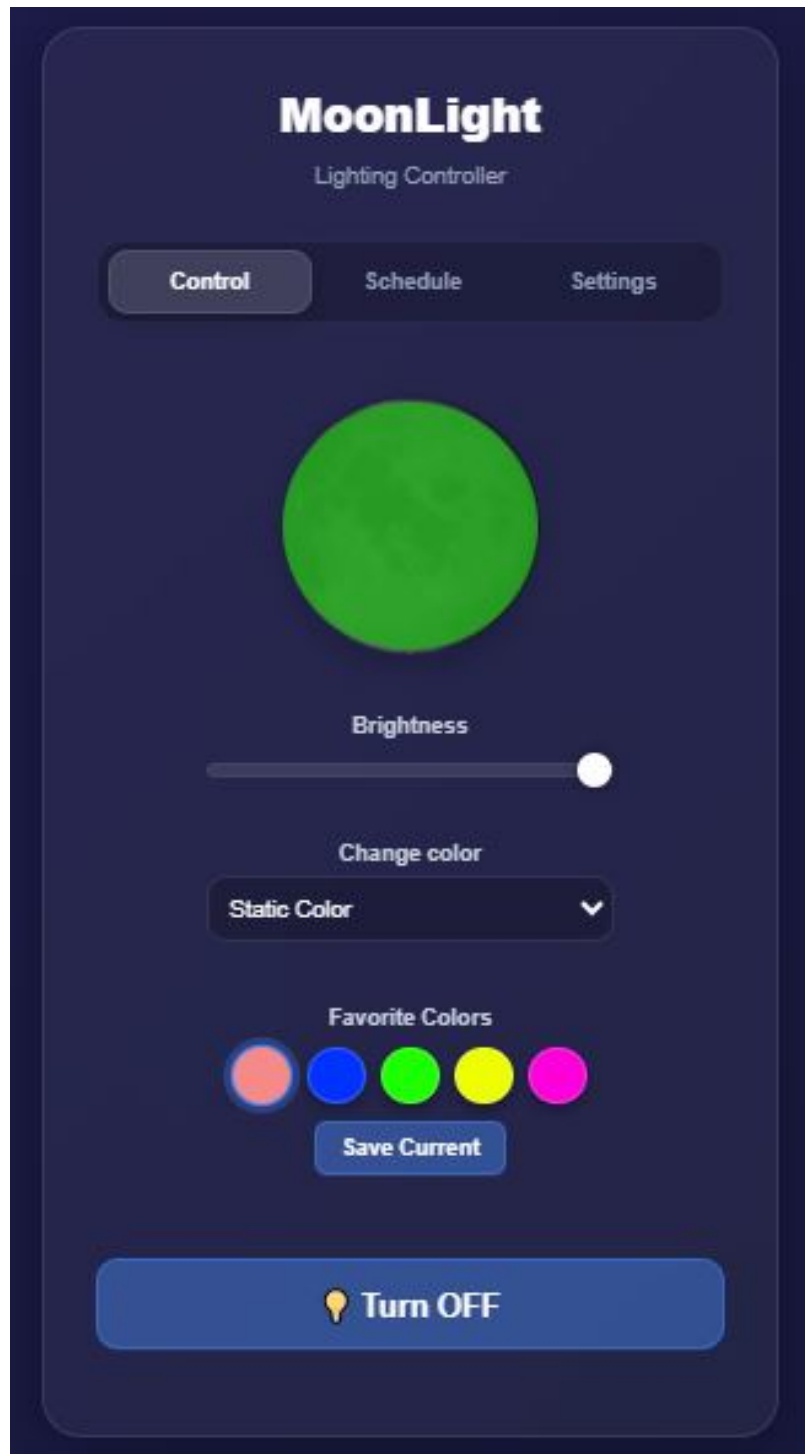


Figure 1: Main Control tab showing the color preview, brightness slider, animation mode selector, favorite color swatches, and power toggle button.

4.1 Color Picker

A large circular color input element is presented over a thematic rotating moon animation. Clicking or dragging within the color wheel selects the desired hue and saturation. The selected color is transmitted to the controller using efficient throttling (maximum update rate ~6–7 Hz) to balance responsiveness with network efficiency. A smooth 250 ms cross-fade animation is rendered locally in the browser for visual continuity.

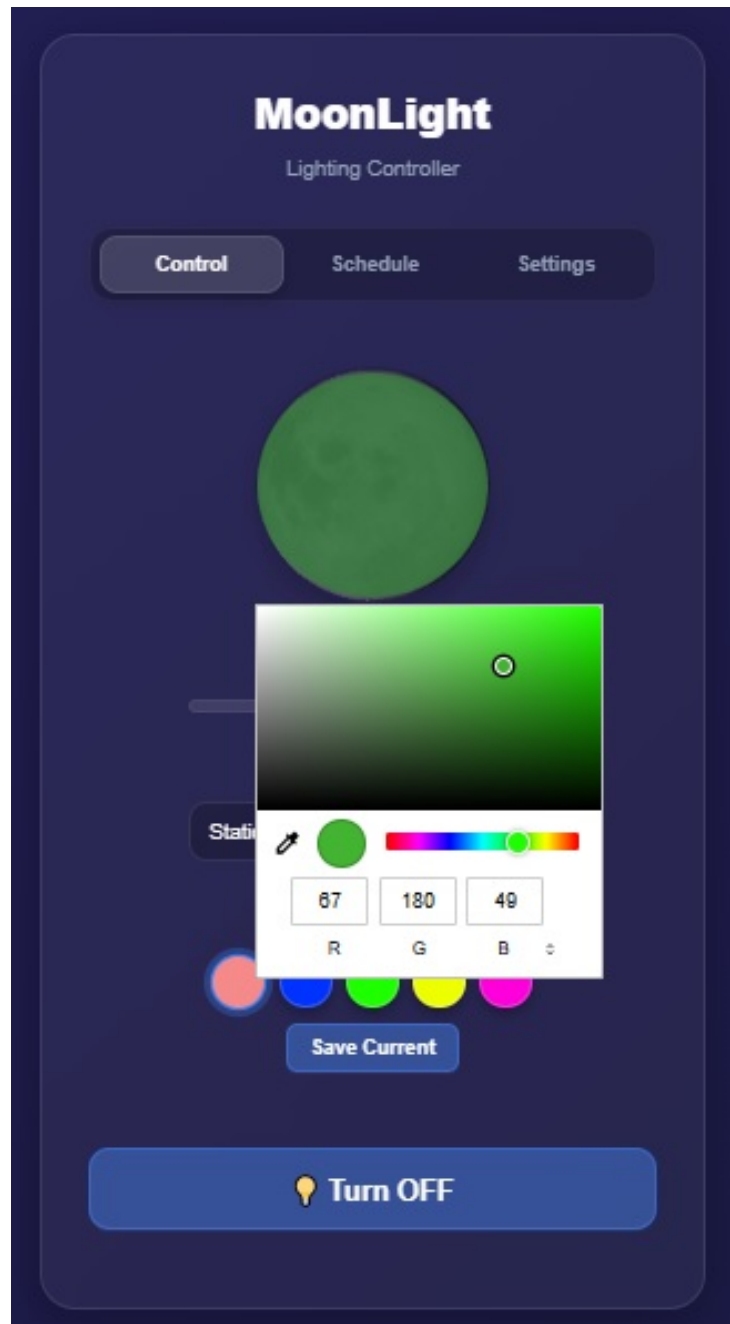


Figure 2: Color picker interface with live RGB value display (67, 180, 49 in example), favorite color presets, and save functionality.

4.2 Brightness Slider

A horizontal slider ranging from 1 % to 100 % scales the PWM duty cycle of all three color channels proportionally. The current brightness value is overlaid temporarily on the color picker when adjusted.

Brightness changes are also throttled and persisted automatically.

4.3 Animation / Scroll Modes

The "Change color" dropdown offers seven operating modes for the LEDs:

- **Static Color** — LEDs remain at the user-selected color and brightness indefinitely.
- **Scroll (Smooth)** — Continuous, smooth RGB color cycling generated by three phase-shifted sine waves. Produces a pleasing rainbow-like transition without abrupt changes.
- **Every 1 minute / 5 minutes / 15 minutes / 1 hour / 1 day** — At the chosen interval, the controller generates a new random RGB color and applies it. Ideal for long-term dynamic lighting effects or seasonal ambiance.

4.4 Favorite Colors

Five circular swatches are provided for rapid recall of frequently used colors. Clicking any swatch immediately applies that color (and current brightness) to the LEDs. The currently selected swatch is highlighted with a blue border. To store the active color into a slot, first select the target slot (it becomes highlighted), then click the **Save Current** button. Favorites are persisted across reboots.

4.5 Power Toggle

A prominent button at the bottom of the Control tab toggles the LED output between fully on (respecting current color/brightness) and fully off. The power state is saved to flash and respected by scheduling logic and cloud commands. The button label and color update dynamically to reflect the current state.

5. Web User Interface — Schedule Tab

The Schedule tab enables fully automatic power management without requiring external timers or smart-home hubs. Three mutually exclusive modes are available.

5.1 Schedule Modes

- **Disabled** — Automation is turned off. The lamp responds only to manual control, cloud commands, and the physical power toggle.
- **Custom Times** — User-specified fixed ON and OFF times (24-hour HH:MM format). The controller evaluates the current local time every 30 seconds and issues the appropriate power command when a transition point is reached.
- **Sun Sync (Sunrise/Sunset)** — The controller automatically calculates daily ON (sunset) and OFF (sunrise) times for the configured geographic location. This mode requires periodic internet access to fetch accurate solar data.

5.2 Sun Sync Configuration Parameters

- **Latitude / Longitude** — Geographic coordinates in decimal degrees. On first internet connection the device attempts to auto-detect location via ip-api.com; values may be manually overridden for precise control or multi-location use.
- **Timezone (UTC Offset in Hours)** — Local time offset from UTC (supports half-hour increments, e.g., 5.5 for India Standard Time).
- **Sunrise Shift / Sunset Shift (Minutes)** — Integer offsets applied to the astronomical sunrise and sunset times. Negative values move the trigger earlier; positive values move it later. Example: Sunset Shift = +30 turns the lights on 30 minutes after actual sunset.

5.3 Time Synchronization and Behavior

Accurate local time is obtained via NTP (pool.ntp.org) upon successful Wi-Fi association. The Schedule engine re-evaluates every 30 seconds. When a scheduled transition occurs, the new power state is written to flash and the LED output is updated accordingly. In Sun Sync mode, solar times are refreshed approximately once per day (or on demand after parameter changes) to automatically accommodate seasonal variations and daylight-saving time changes.

***Note:** If the device loses internet connectivity for an extended period, NTP time may drift and Sun Sync calculations will become stale. A stable internet connection is strongly recommended when using automated scheduling.*

The screenshot displays the MoonLight Lighting Controller interface. At the top, the title "MoonLight" is prominently displayed in white, with "Lighting Controller" in a smaller font below it. A navigation bar contains three tabs: "Control", "Schedule" (which is highlighted), and "Settings".

Under the "Schedule" tab, the "Automation" section is active. It features a "Schedule Mode" dropdown menu currently set to "Sun Sync (Sunrise/Sunset)". Below this, a descriptive text states: "Sun Sync uses the internet to fetch local sunset (ON) and sunrise (OFF) times."

Two input fields for shift times are present: "Sunrise Shift (Mins)" with a value of 30, and "Sunset Shift (Mins)" with a value of -45. A small icon next to the sunset shift field indicates a toggle between before and after sunset. Below these, a note specifies "[-] before, [+] after."

Geographic coordinates are entered in two fields: "Latitude (e.g. 40.71)" with the value 53.5815, and "Longitude (e.g. -74.00)" with the value -2.1778.

The "Timezone (UTC Offset in Hours)" field is set to 1.

A large blue "Save Schedule" button is positioned at the bottom of the configuration area. Below this button, the status "Currently Active ON: 18:00 | OFF: 06:30" is displayed.

Figure 5: Sun Sync (Sunrise/Sunset) automation configuration with shift offsets, geographic coordinates, timezone, and resulting active ON/OFF times.

6. Web User Interface — Settings Tab

The Settings tab is organized into three sub-tabs providing network, cloud, and system-level configuration options.

6.1 WiFi Sub-tab

- Displays real-time connection status and the currently associated SSID when in Station Mode.
- **Forget WiFi** button clears stored credentials and forces an immediate reboot into AP Mode.
- A secondary form allows entry of a new SSID/password pair to switch networks without performing a factory reset.

6.2 Alexa / Google (Cloud) Sub-tab

Integration with Amazon Alexa and Google Home is achieved through the free Sinric Pro cloud service (<https://sinric.pro>). The free tier permits up to three devices per account.

After creating a Sinric Pro account and adding a "Smart Light Bulb" device, copy the generated **App Key**, **App Secret**, and **Device ID** into the provided form fields and click **Save Cloud Settings**. The controller will reboot and establish a persistent WebSocket connection to the Sinric Pro broker. Once connected, the following voice commands are supported:

- "Alexa, turn MoonLight on / off"
- "Alexa, set MoonLight brightness to 50 percent"
- "Alexa, set MoonLight color to warm white / blue / purple"
- Equivalent Google Home commands are also supported.

A convenient step-by-step guide for obtaining the required credentials is accessible via the "Need help finding these keys?" link, which opens an on-screen modal dialog.

MoonLight
Lighting Controller

Control Schedule **Settings**

WiFi **Alexa/Google** System

Cloud Integration (Alexa / Google)

[Need help finding these keys?](#)

App Key
e.g. de0bxxxx...

App Secret
e.g. 5f36xxxx...

Device ID
e.g. 5dc1xxxx...

Save Cloud Settings

Figure 3: Sinric Pro cloud integration form for entering App Key, App Secret, and Device ID to enable Alexa and Google Home control.

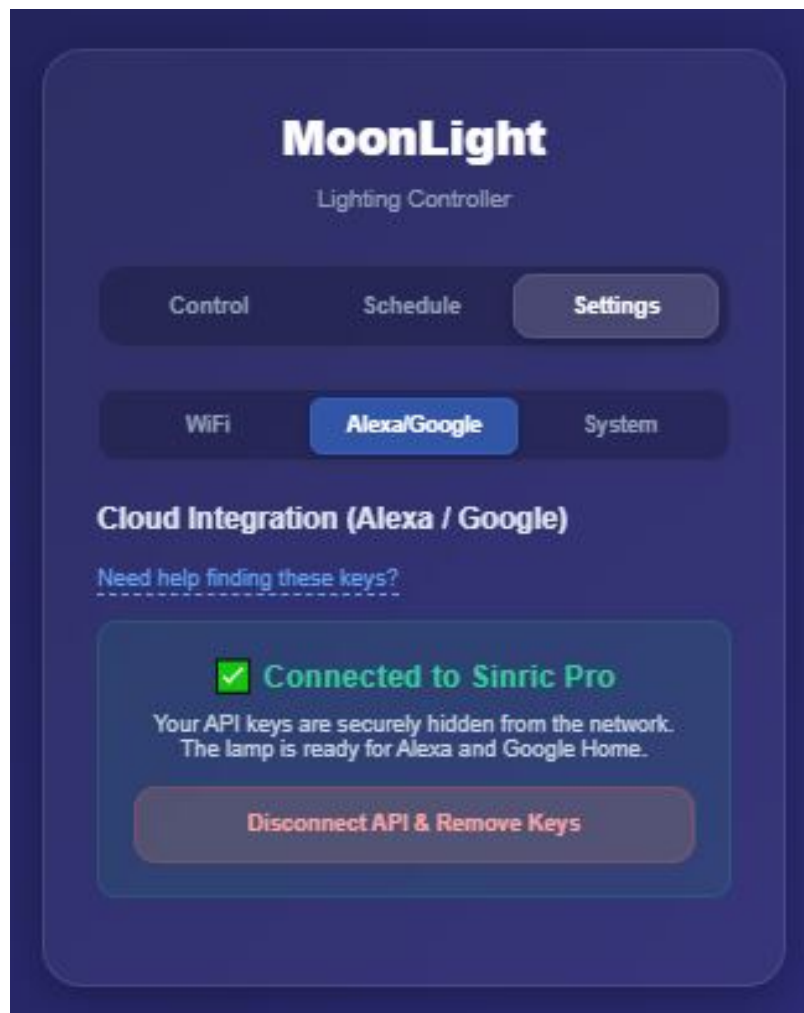


Figure 4: Successful connection status screen confirming the lamp is ready for Alexa and Google Home voice commands.

6.3 System Sub-tab

Firmware Update Section

- Current firmware version is displayed for reference.
- **Check for Updates** queries the remote update server and reports availability.
- When an update is available, an **Install Update Now** button appears. A confirmation dialog warns that the device will be unresponsive for approximately one minute.
- **Notify via LEDs** — When enabled, the controller periodically enters a green pulsing alert state (once per minute) to visually indicate that new firmware is available on the server.
- **Install updates automatically** — Enables silent, scheduled OTA updates within a user-defined hourly window (e.g., 03:00–04:00). The device must be powered on and internet-connected during the window for the update to occur.

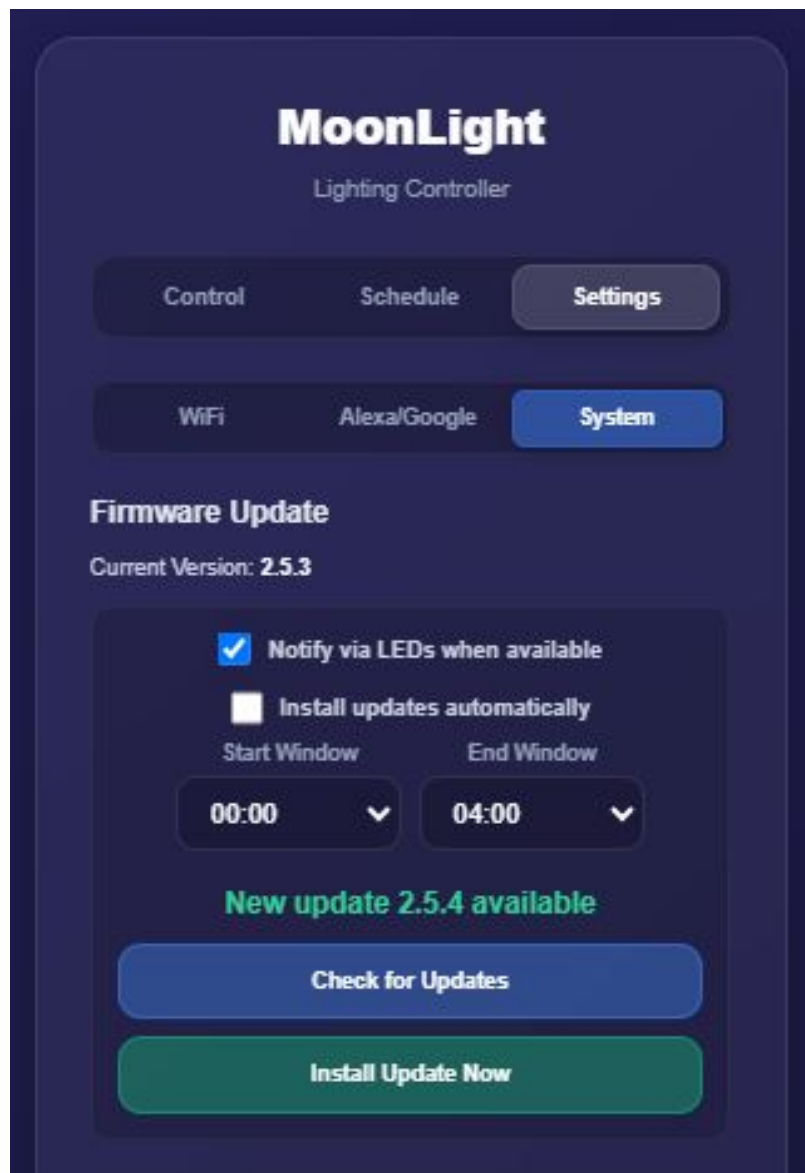


Figure 6: Firmware update interface showing current version, new version availability notification, auto-update scheduling options, and manual check/install buttons.

Advanced Settings

- **Enable serial debug logs** — Toggles verbose diagnostic output on the USB serial console (115200 baud). Useful for advanced troubleshooting.
- **Factory Reset** — Clears all user-configurable data (Wi-Fi credentials, colors, favorites, schedule parameters, cloud keys, and preferences). The device restarts in AP Mode ready for re-setup.

Warning: Factory reset is irreversible via the web interface. All customizations will be lost. All personal settings will be cleared.

7. LED Status Indicators and Animations

In addition to providing illumination, the RGB LED output serves as a visual status indicator during boot, network transitions, and firmware events. The internal state machine prioritizes status animations over normal lighting when appropriate.

State	Visual Behavior	Condition / Meaning
AP_BLINK	Blue blinking (500 ms cycle)	Access Point Mode active — awaiting configuration
CONNECTING_FADE	Smooth rainbow color fade (slow sine interpolation)	Attempting association with home Wi-Fi network
CONNECTED_BLINK	Three rapid green blinks, then return to NORMAL	Successful Wi-Fi connection established
NORMAL	Static color, smooth scroll, or interval random colors (per user setting)	Normal runtime operation; user or schedule control active
UPDATE_ALERT	Green pulse / dim effect (2 s duration, repeats every 60 s if notify enabled)	New firmware version detected on update server

When the lamp is powered off via the UI, cloud command, or schedule, all LED channels are driven to 0 % duty cycle except during the UPDATE_ALERT state (which may still produce a visible green pulse even when the lamp is nominally off).

8. Over-the-Air Firmware Updates

The MoonLight controller supports secure, user-initiated or fully automatic over-the-air firmware updates. New firmware images are hosted on the manufacturer's update infrastructure (cloudapps.zapto.org).

8.1 Manual Update Process

1. Navigate to **Settings** → **System** and click **Check for Updates**.
2. If a newer version is available, the remote version number and an **Install Update Now** button are displayed.
3. Click the install button and confirm the warning dialog. The controller will download the binary, erase the inactive OTA partition, flash the new image, and reboot automatically.
4. Allow approximately 60–90 seconds for the complete process. The web interface will become unreachable during this window.

8.2 Automatic (Silent) Updates

When the "Install updates automatically" checkbox is enabled, the controller performs a version check every 60 seconds while connected. If an update is pending and the current local time falls within the configured hourly window, the update is applied silently without user intervention. This feature is ideal for installations where physical access is limited.

Caution: Do not remove power from the controller while an update is in progress. Although the ESP32 OTA mechanism is robust, an interrupted flash may require recovery via USB serial programming tools.

9. Voice Assistant Integration (Sinric Pro)

Once Sinric Pro credentials are configured, the MoonLight appears as a standard smart light bulb within the Alexa or Google Home ecosystem. The following capabilities are exposed:

- Power state (ON / OFF)
- Brightness level (0–100 %)
- Color (full RGB via voice color names or hex values where supported)

When a color or brightness command is received from the cloud while an animation mode is active, the animation is automatically cancelled and the device switches to static color mode to honor the voice command. Subsequent manual interaction via the web UI can re-enable animations.

10. Troubleshooting Guide

Issue: Device remains in AP Mode after entering Wi-Fi credentials

Resolution: Verify that the SSID and password are entered exactly (case-sensitive). Confirm that the network is 2.4 GHz. Check router logs for association attempts. Re-enter credentials via 192.168.4.1.

Issue: mDNS name (MoonLight_SN.local) does not resolve

Resolution: Not all routers or client devices support mDNS reliably. Use the device's IP address (obtainable from router DHCP lease table or serial console) instead.

Issue: Voice commands have no effect

Resolution: Confirm that the three Sinric Pro credentials are correctly entered and that the device shows "Connected to Sinric Pro" in the UI. Restart the controller after saving new credentials. Verify internet connectivity and that the Sinric Pro account is active.

Issue: Schedule does not trigger at expected times

Resolution: Ensure the device has internet access for NTP synchronization. In Sun Sync mode, verify latitude, longitude, and timezone offset are correct. Check the "Currently Active ON/OFF" display at the bottom of the Schedule tab for the effective times being used.

Issue: Firmware update fails or hangs

Resolution: Check internet connection stability. Large firmware images require sustained throughput. Review serial debug output (enable in UI) for detailed HTTP and OTA progress messages. Retry the update later if the server is temporarily unavailable.

Issue: LEDs do not light or show incorrect colors

Resolution: Confirm that the physical LED strip or module is wired to the correct GPIO pins (Red=GPIO3, Green=GPIO4, Blue=GPIO2) and that a suitable current-limiting driver or power supply is used. Verify brightness is not set to 0 % and that the power state is ON.

11. Technical Specifications and Limitations

Parameter	Specification
Firmware Version	2.5.4
Microcontroller	ESP32 (Xtensa dual-core LX6)
Wi-Fi Standard	802.11 b/g/n (2.4 GHz only)
LED Channels	3× PWM (8-bit resolution, ~1 kHz carrier)
PWM Pins	GPIO 2 (Blue), GPIO 3 (Red), GPIO 4 (Green)
Persistent Storage	ESP32 NVS (Preferences partition)
Update Mechanism	HTTPUpdate over HTTPS (insecure mode for compatibility)
Time Source	NTP (pool.ntp.org) + optional solar API
Cloud Platform	Sinric Pro (WebSocket)
Maximum LED Current	Determined by external driver / power supply (controller logic only)
Operating Temperature	0 °C to +50 °C (typical for ESP32 modules)
mDNS Hostname	MoonLight_SN.local or MoonLight.local

Known Limitations

- 5 GHz Wi-Fi networks are not supported.
- High-power LED loads require external MOSFET drivers or constant-current supplies; the ESP32 GPIO cannot source significant current directly.
- OTA updates require sufficient free RAM and a stable internet connection; very large images may fail on memory-constrained configurations.
- Sun Sync calculations assume a clear horizon; local obstructions (buildings, trees) may cause minor timing discrepancies versus actual observed sunrise/sunset.

12. Safety, Maintenance, and Support

12.1 Safety Precautions

- Always match the LED load current and voltage requirements to an appropriately rated power supply and driver circuitry.
- Never interrupt power during an active firmware update.
- The device is intended for indoor, dry-location use only.

- Factory reset erases all configuration data. The device returns to its initial setup state.

12.2 Maintenance

No routine maintenance is required beyond keeping the device powered and, for optimal scheduling and update functionality, maintaining internet connectivity. Periodically verify that the latest firmware is installed via the Check for Updates function.

12.3 Support

For technical assistance, enable serial debug logging in the web interface and capture console output during the observed issue. The device also transmits periodic anonymized status reports (color, brightness, schedule mode, RSSI, firmware version) to the backend server when a Serial Number has been assigned; these reports assist remote diagnostics.

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